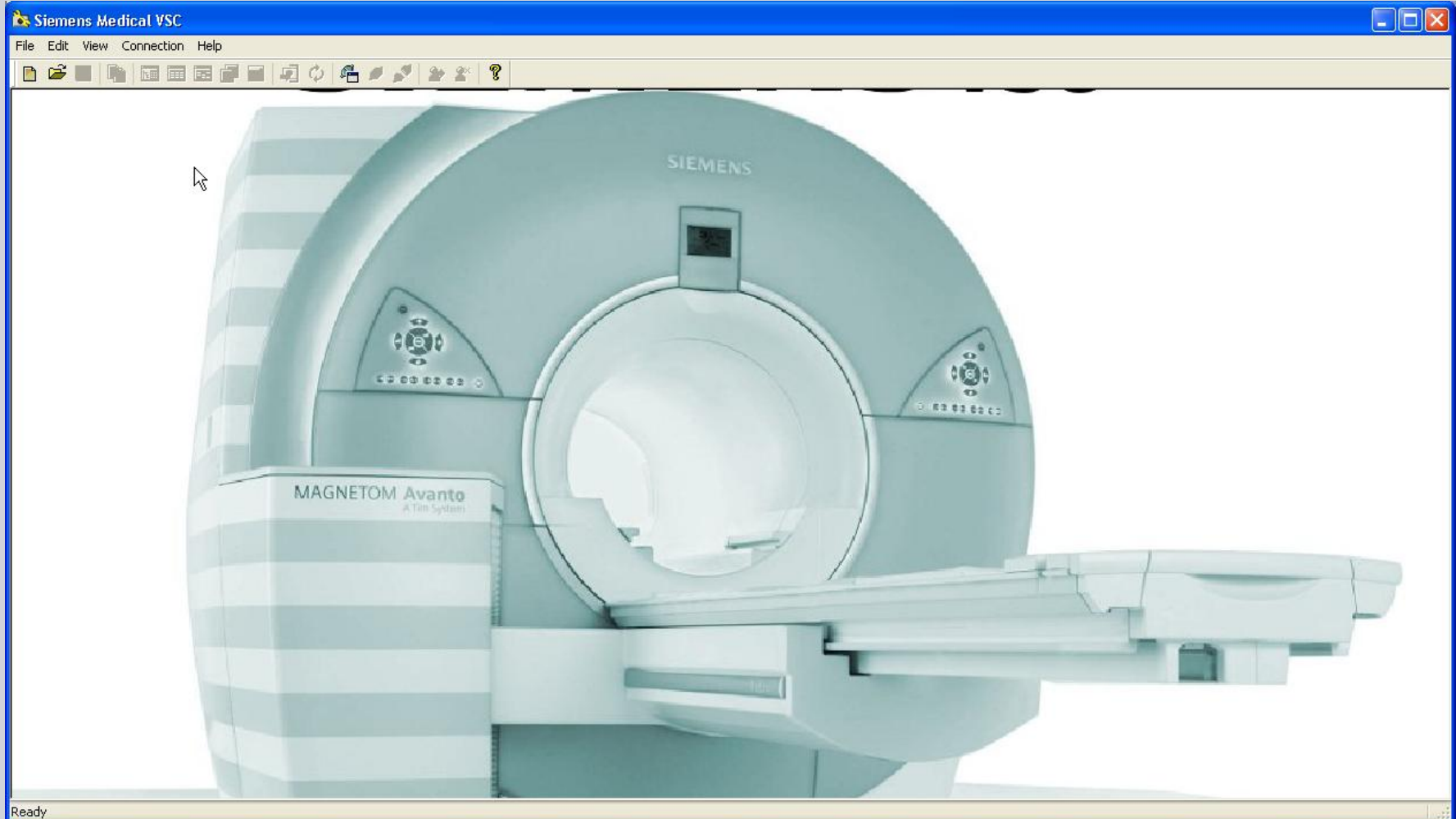


Siemens Medical VSC – Software for the VLM device (Gradient Supervision)

SIEMENS



For internal use only

Software Installation

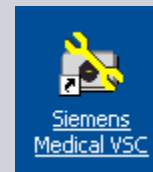
Note: There have been compatibility problems with the VLM Software in combination with Win 7.
The current version is fully compatible to Win 7.

1. Download the VLM Software_Win7.ZIP file from the Intranet (Product Information > MR > Downloads > Gradient_Supervision:

<https://bps-healthcare.siemens.com/productinformation/ProductInformation.aspx?uuid=7abf6bb6-5daf-440c-9920-fab2d4640084>

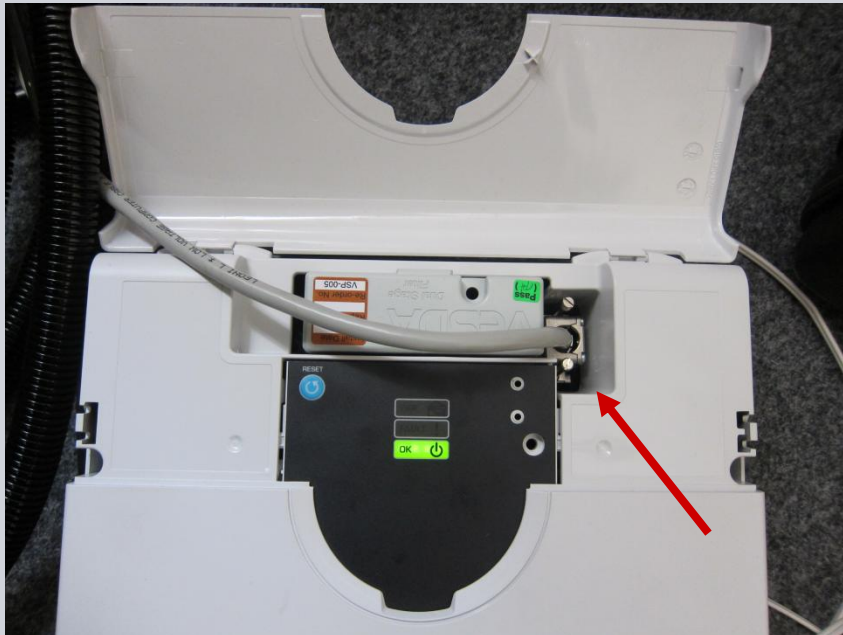
1. Unzip it to your PC and install the SiemensMedicalSetup_30500.msi

2. You will get an Icon on your desktop:



Required Hardware / Connection

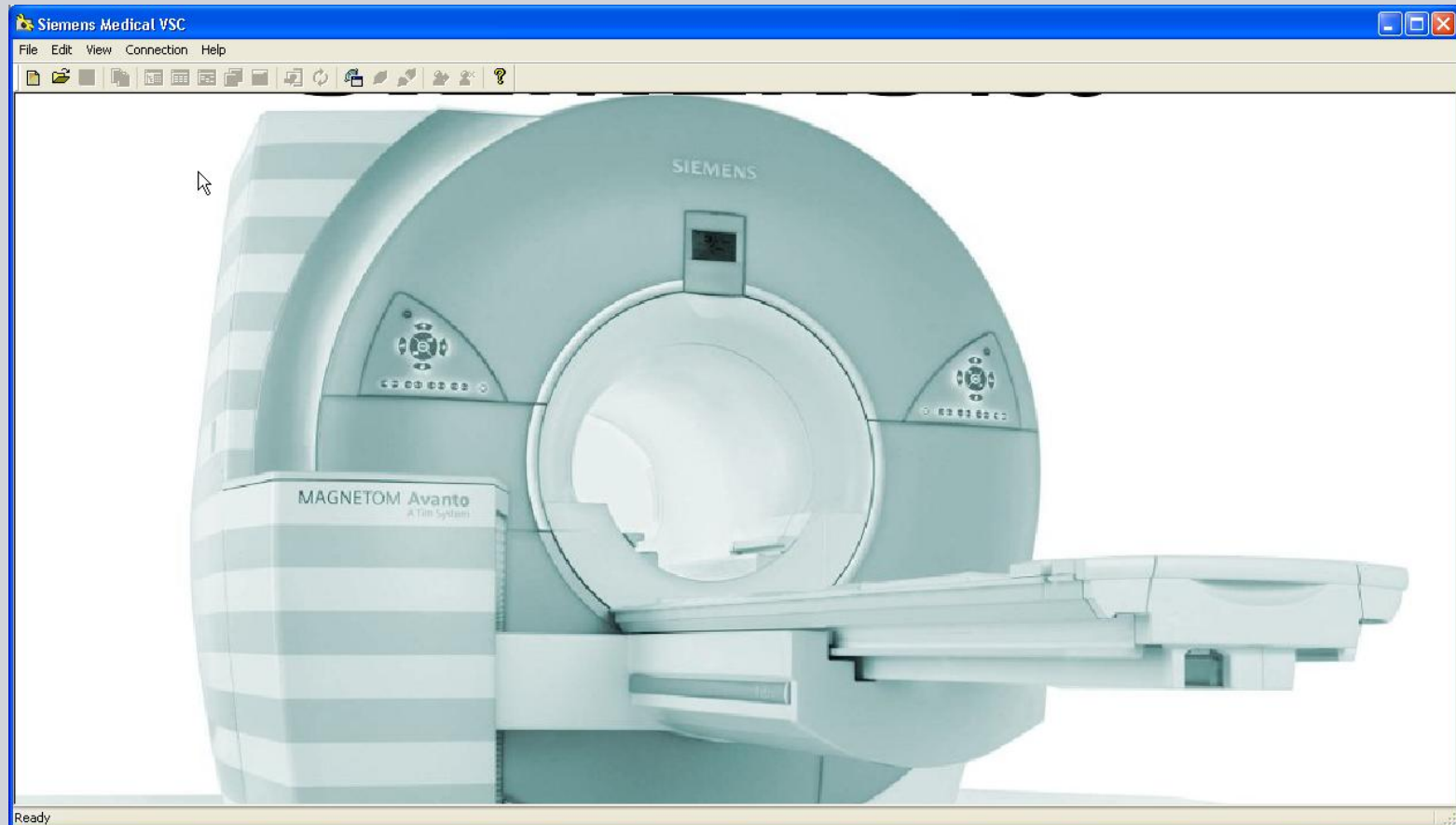
1. RS232 cable 9p-sub-D (female/male, part-no.: 10915402)
2. Laptop with RS232 connector
3. Connect the RS232 cable (male connector) to the socket on front of the VLM-device (under the top cover).
4. Connect the other end of the cable to your PC.
5. Click on the desktop Icon:



For internal use only

Content

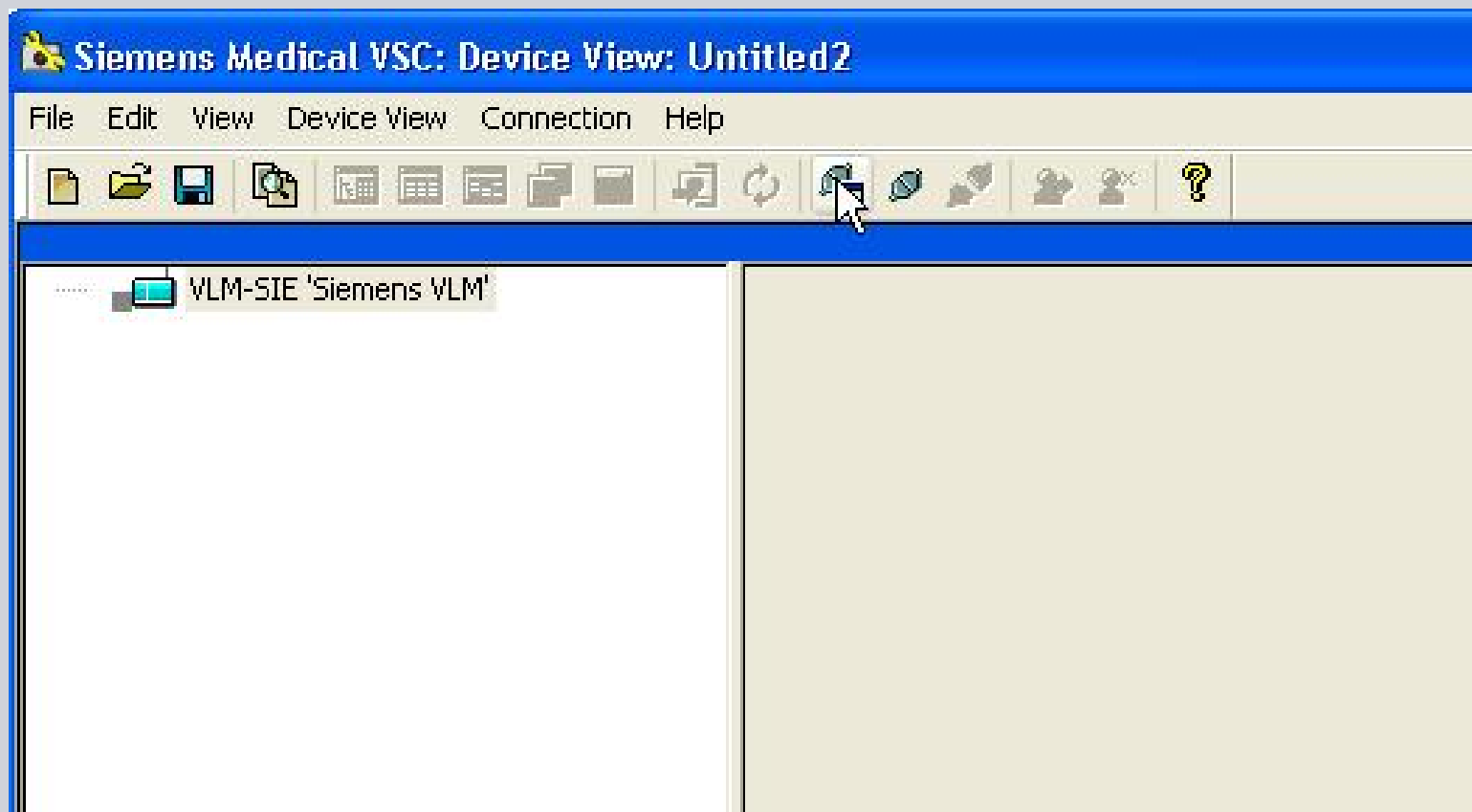
The Software will come up with the VLM user interface



For internal use only

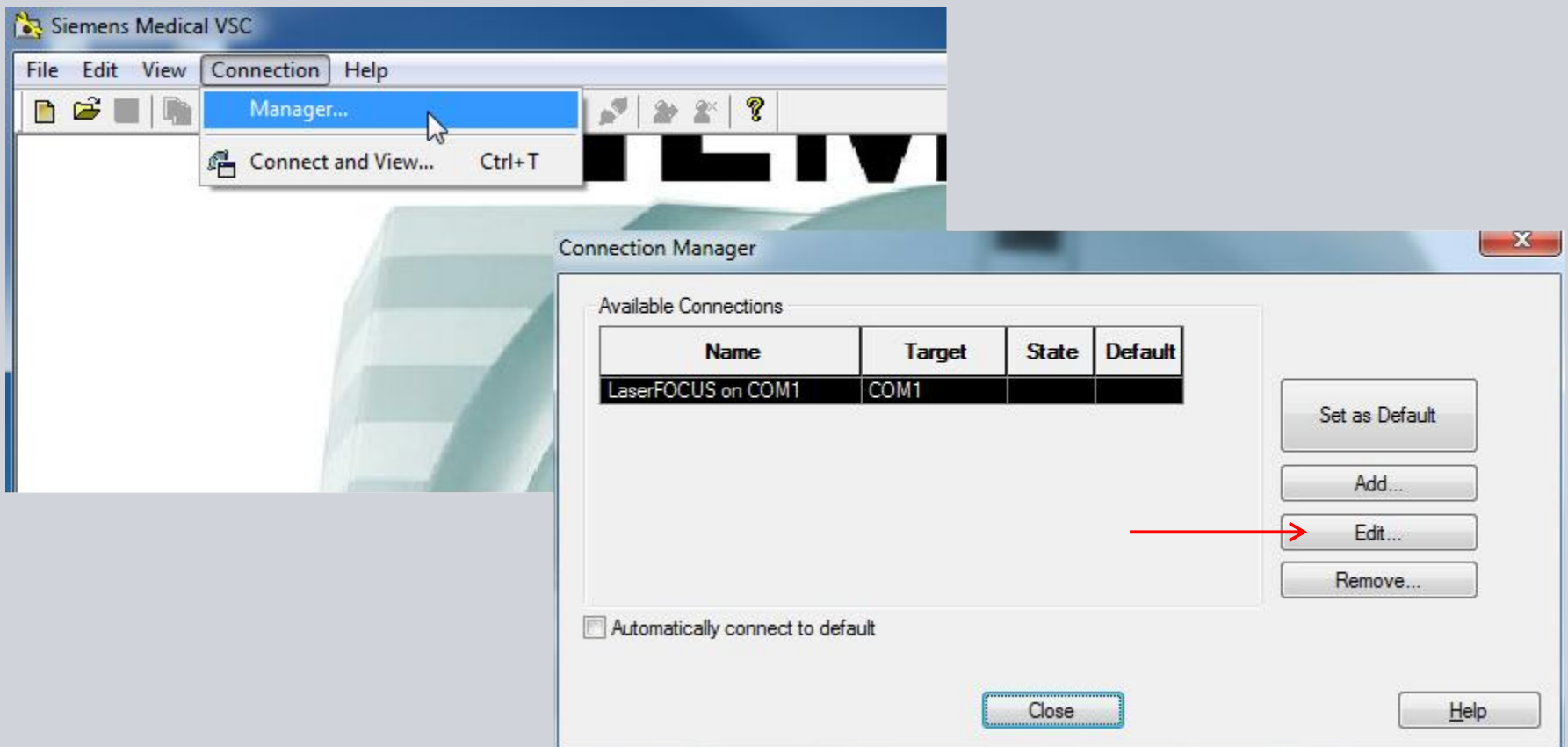
Content

Click the “connect” button in order to start communication with the VLM-device



Communication set-up

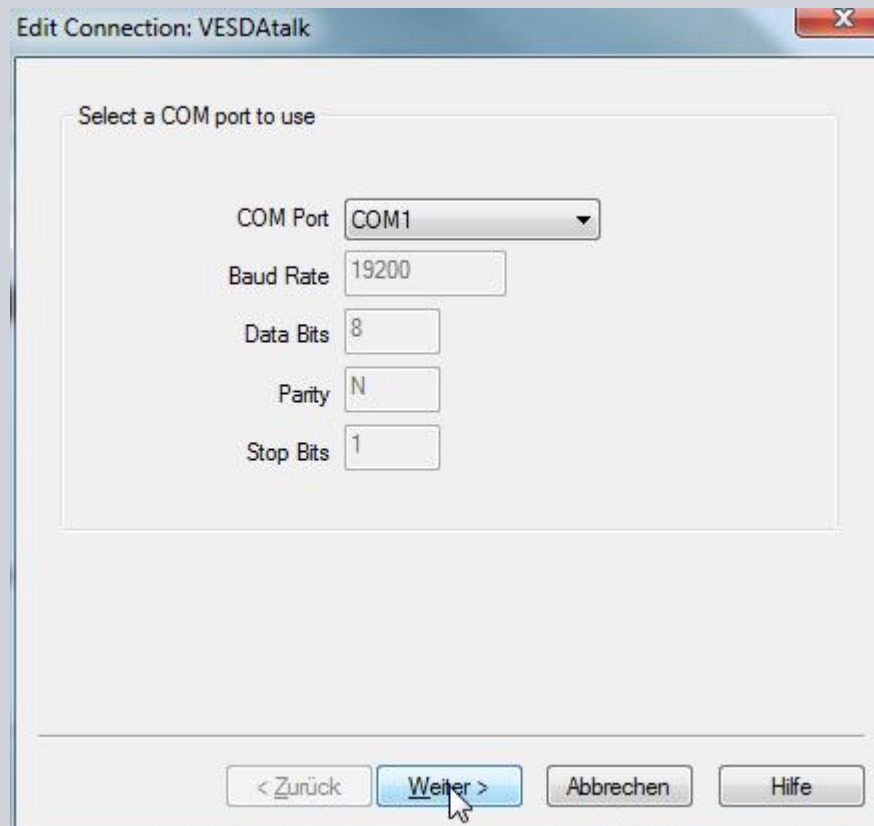
If the communication parameters are not set properly, the communication will fail.
In this case please open the Connection Manager, click on Edit...



For internal use only

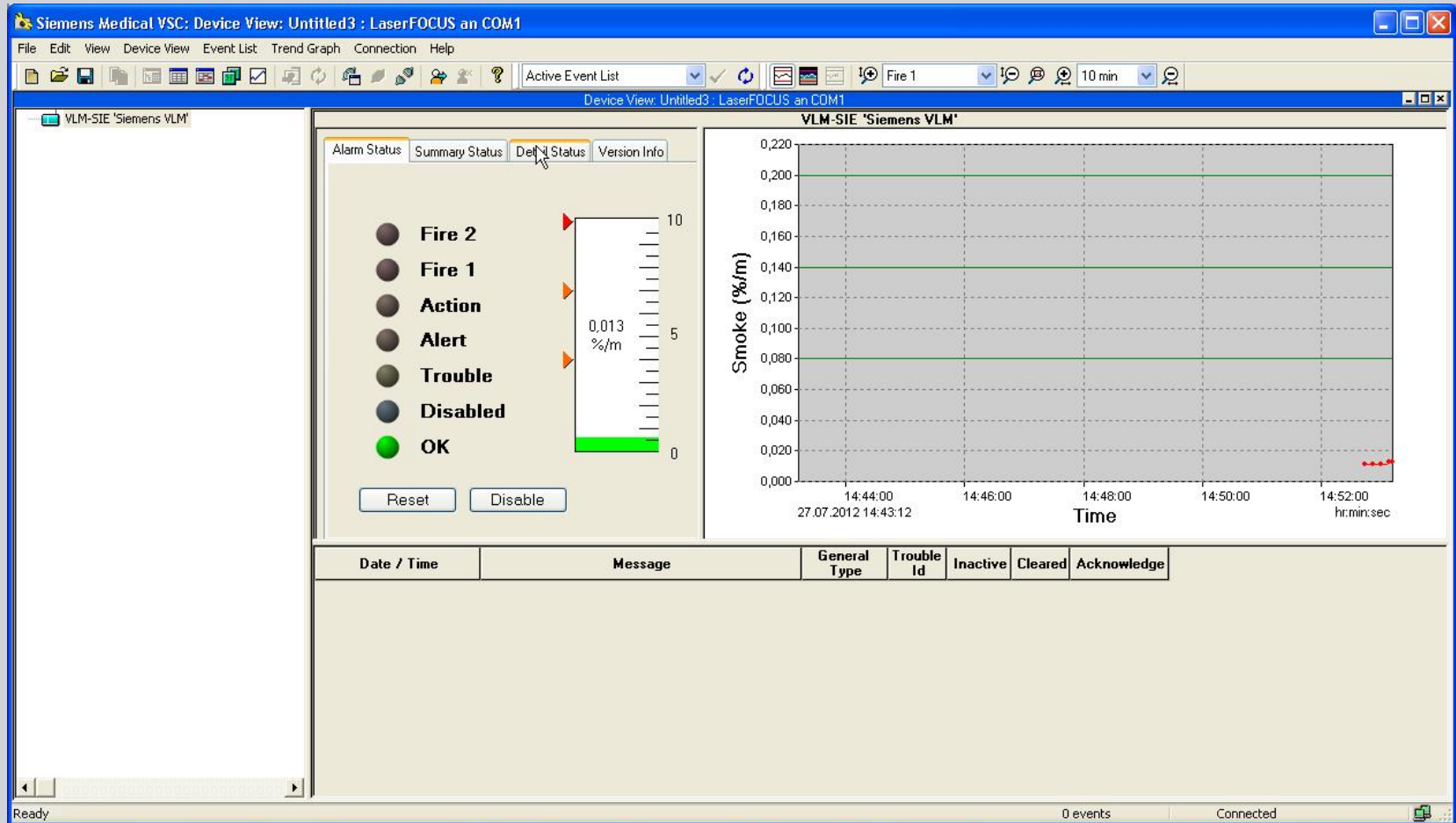
Communication set-up

...and check / correct the communication parameters according to the snapshot below:



User Interface

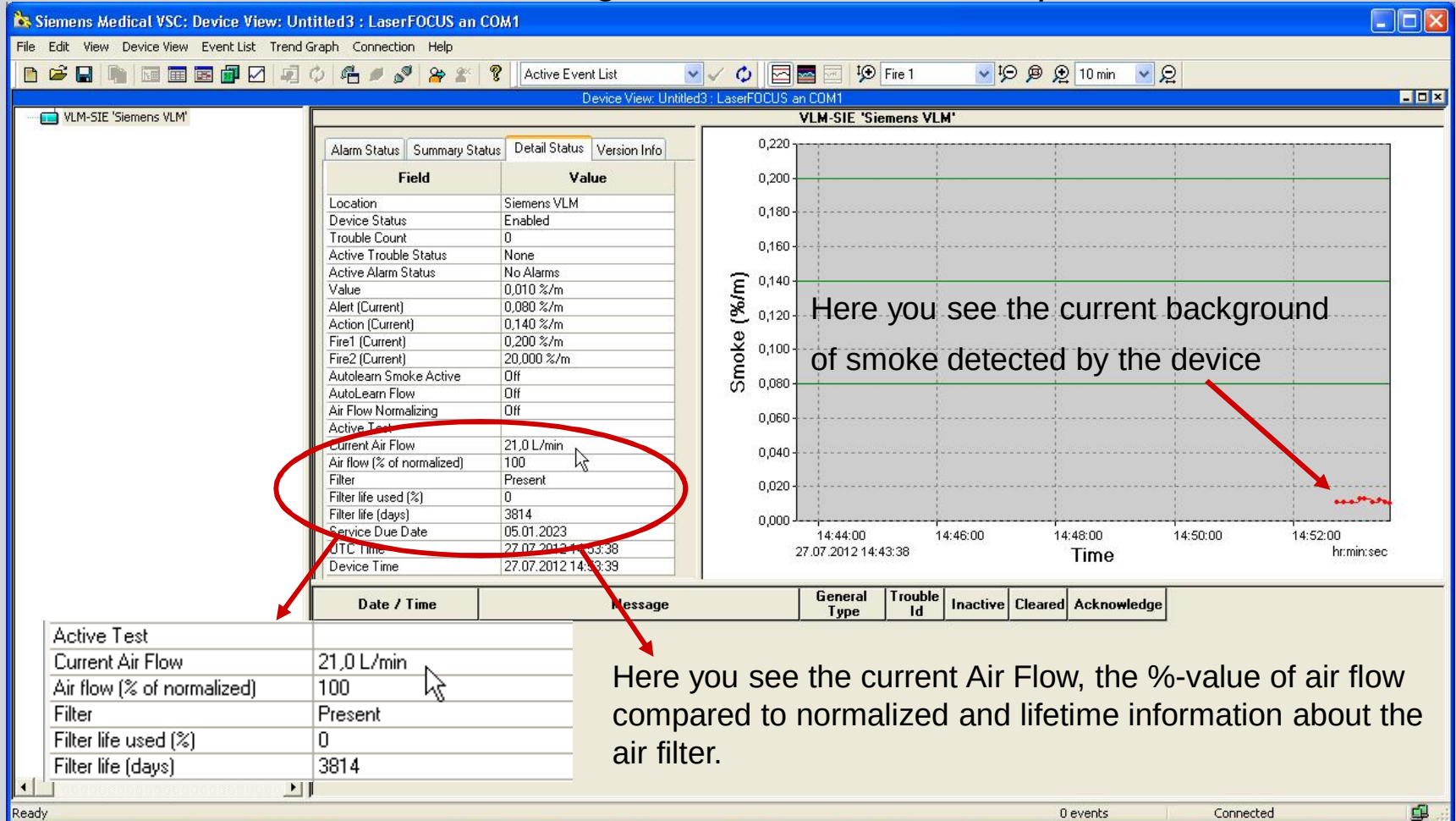
The Device View will open with the Alarm Status and smoke graph



For internal use only

User Interface

Click on the “Detail Status” register to see the most important informations



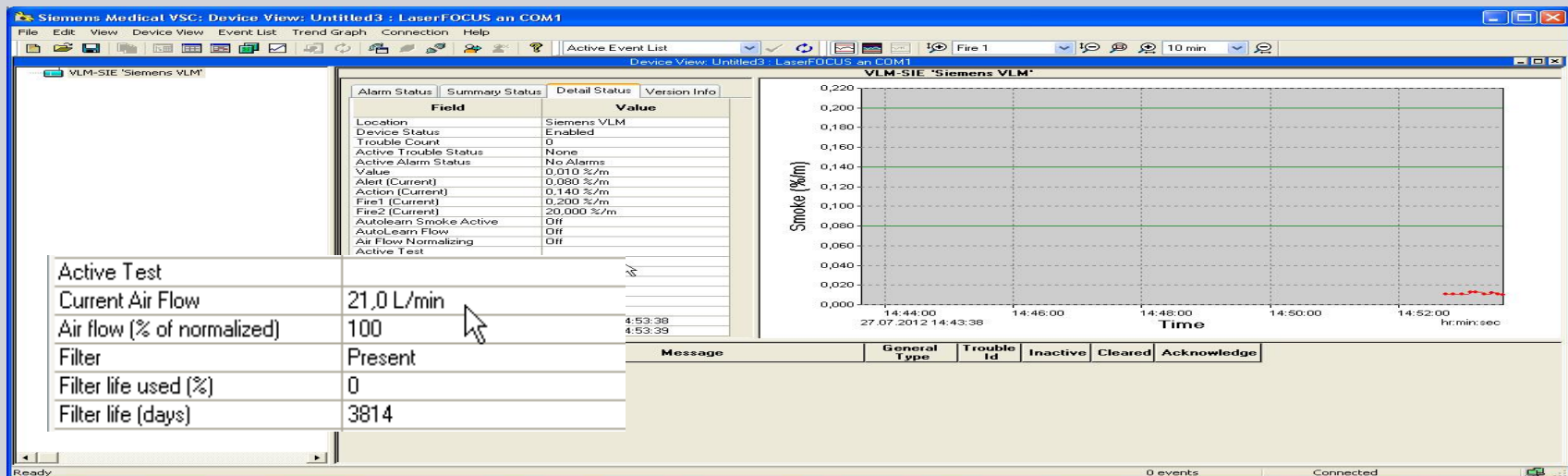
For internal use only

Proper working range of the VLM device

Important Note:

The VLM device does only work properly under the following conditions

- The measured Current Air Flow (see screen shot below) must be in a range between 12 and 24 L/min. Outside this range the VLM device will not work, cannot be normalized and is in FAULT condition.
- The Current Air Flow may deviate for +/- 20% from the normalized 100% value (RF cabin door open/closed). Deviations above the 20% limit will result in a FAULT condition of the VLM device.



For internal use only

Event Log



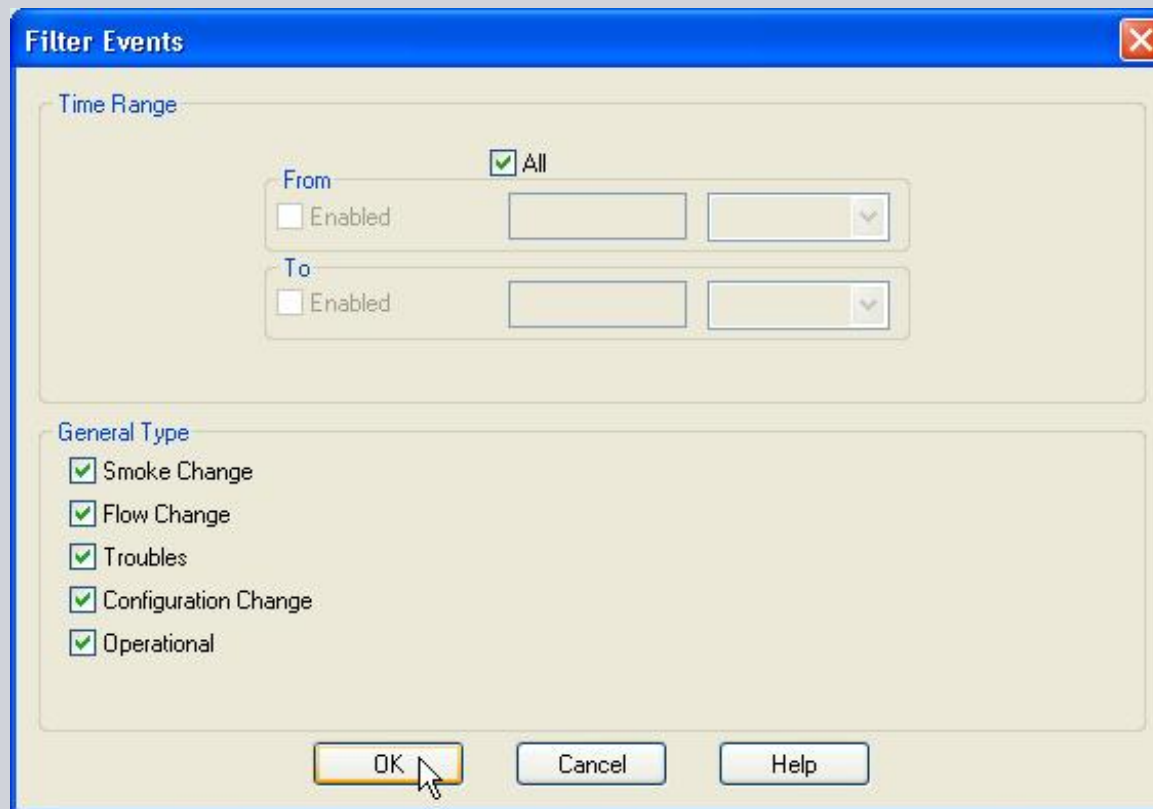
Click on the “Event Log” button to see the list of logged events. This can also be downloaded (file-save as)



The Software will ask you for a 4 digit pin to get access to the Event Log.
Enter 1111 and click OK.

Event Log

Per default all filter events are selected, just say OK.



Filter Events

Time Range

From ☒ All

☐ Enabled

To

☐ Enabled

General Type

☒ Smoke Change

☒ Flow Change

☒ Troubles

☒ Configuration Change

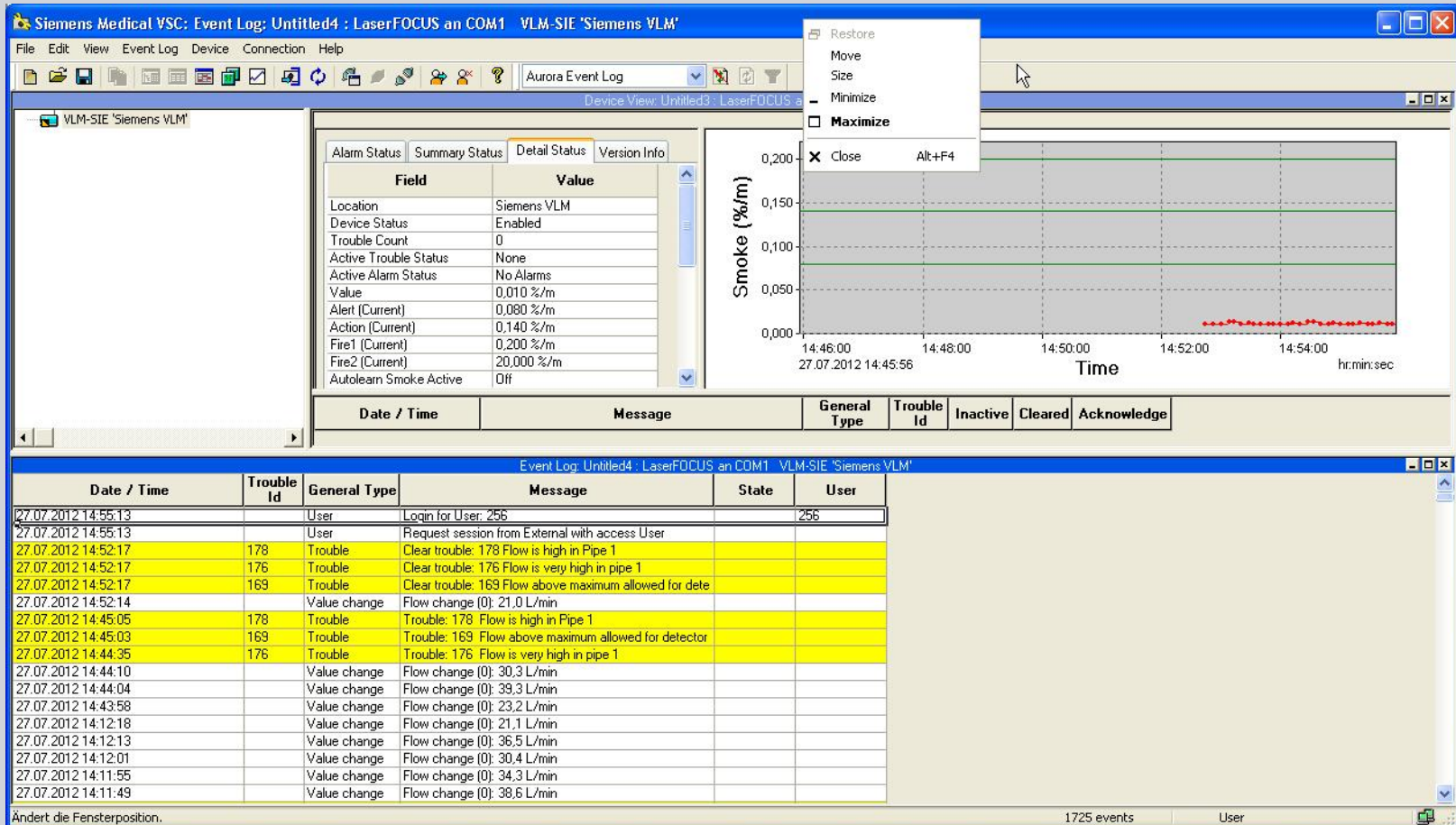
☒ Operational

OK Cancel Help

For internal use only

Event Log

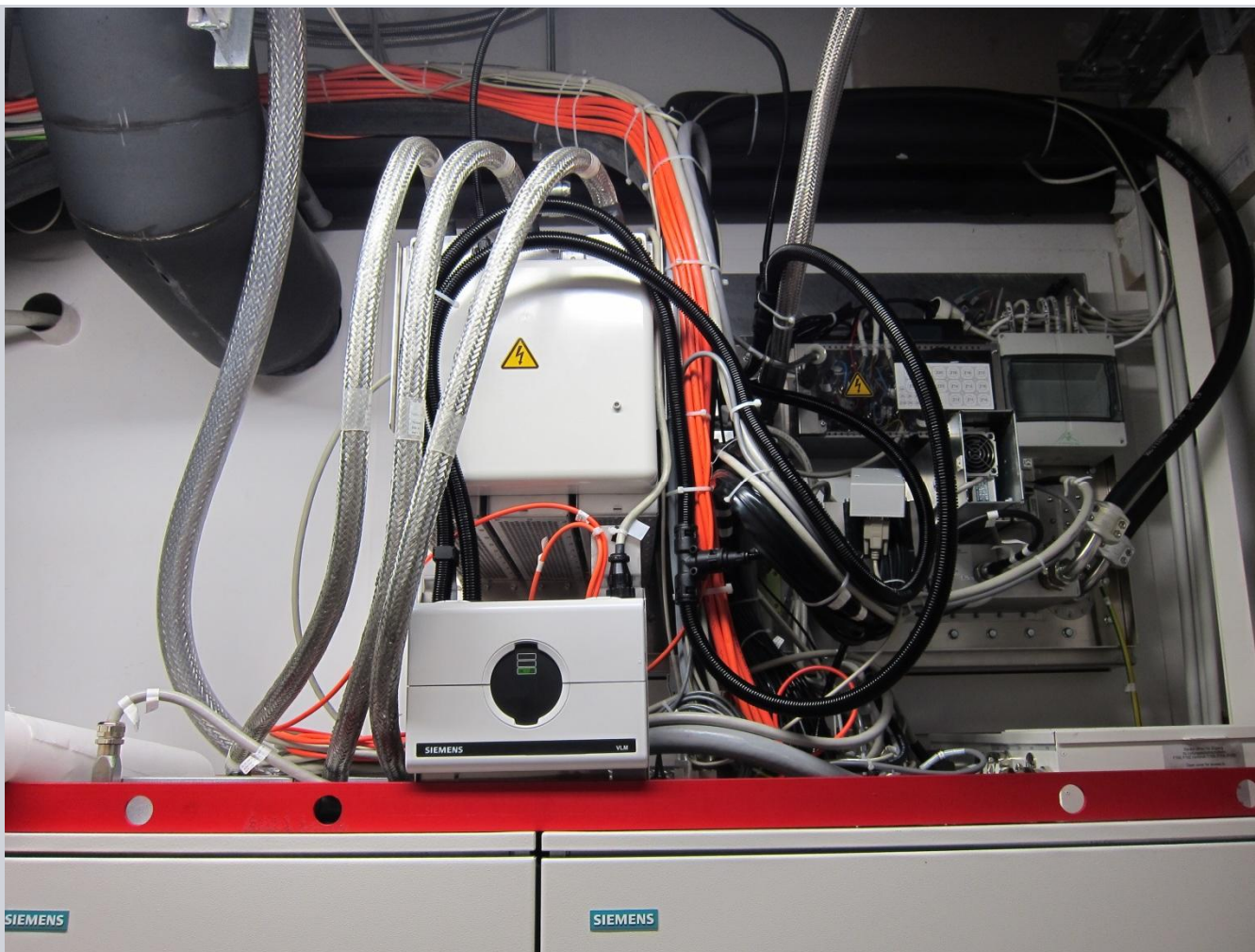
Now you see the Event Log. Yellow events are FAULT conditions and red events are FIRE conditions.



For internal use only

Trouble Shooting – Gradient Supervision

SIEMENS



- 1. Short Introduction to VLM-function**
- 2. Currently seen problems**
- 3. Trouble shooting flow chart**
- 4. Installation of an additional T-piece in the equipment**

VLM-Overview



The VLM-device has only 3 status LEDs

OK (default)

FAULT (flashing, or static – depending on the fault severity – results in a customer pop-up)

FIRE (turns off the GPA modulator immediately and shuts down the system 1min. Later)

The RESET-button is used for resetting FAULTs or normalising the device

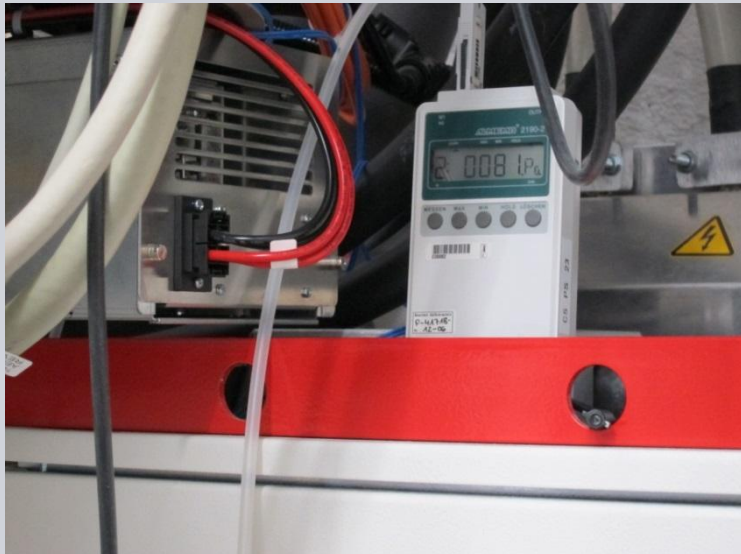


Trouble Shooting

The major problem currently seen in the field is the FAULT LED flashing or being on permanently.

This problem is mainly caused by an incorrect air-flow (Allowed air-flow range between 12 and 24 ltrs./min. for normalisation +/- 20% from normalised value, respectively +/- 30% on devices with serial number >SB8708352).

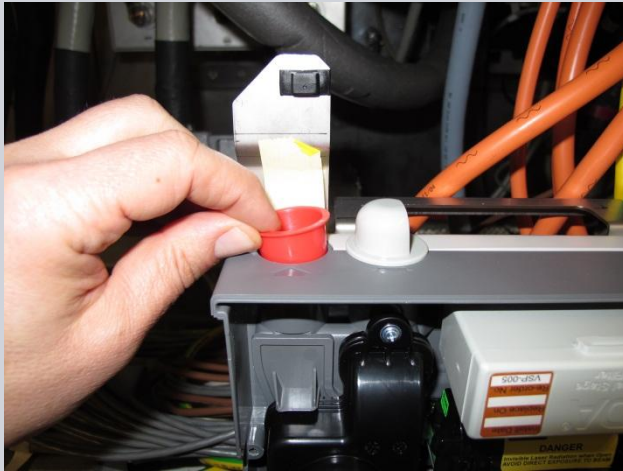
Possible reasons for air-flow problem:



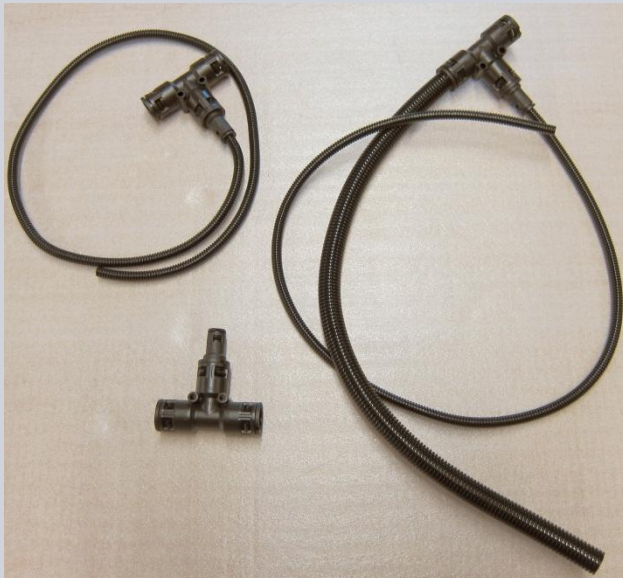
1. Excessive differential pressure between RF-cabin and equipment-room (e.g. Munich: >81Pa). This should be below 50 Pa.

For internal use only

Trouble Shooting



2. Blind cap not removed from VLM-input.
This will result in zero flow and a FAULT condition.



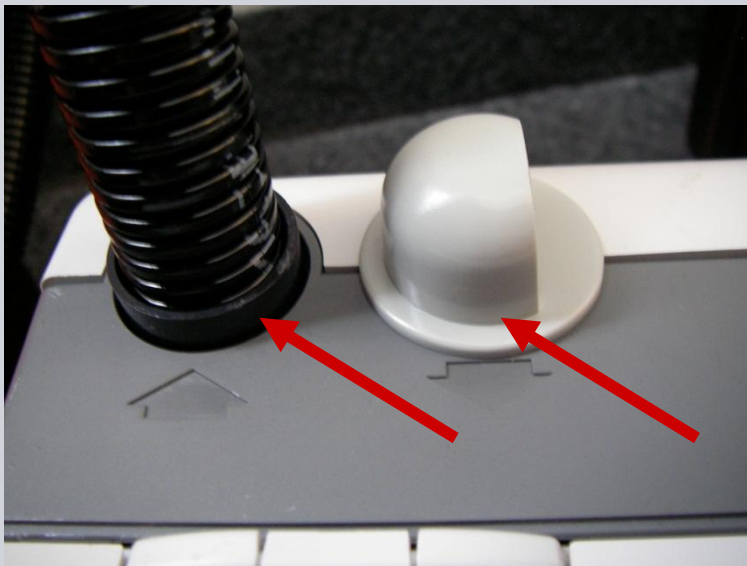
3. Leaking hose connections.
(e.g. when the hose is not properly cut in the groove ground and does not have smooth edges)

For internal use only

Trouble Shooting



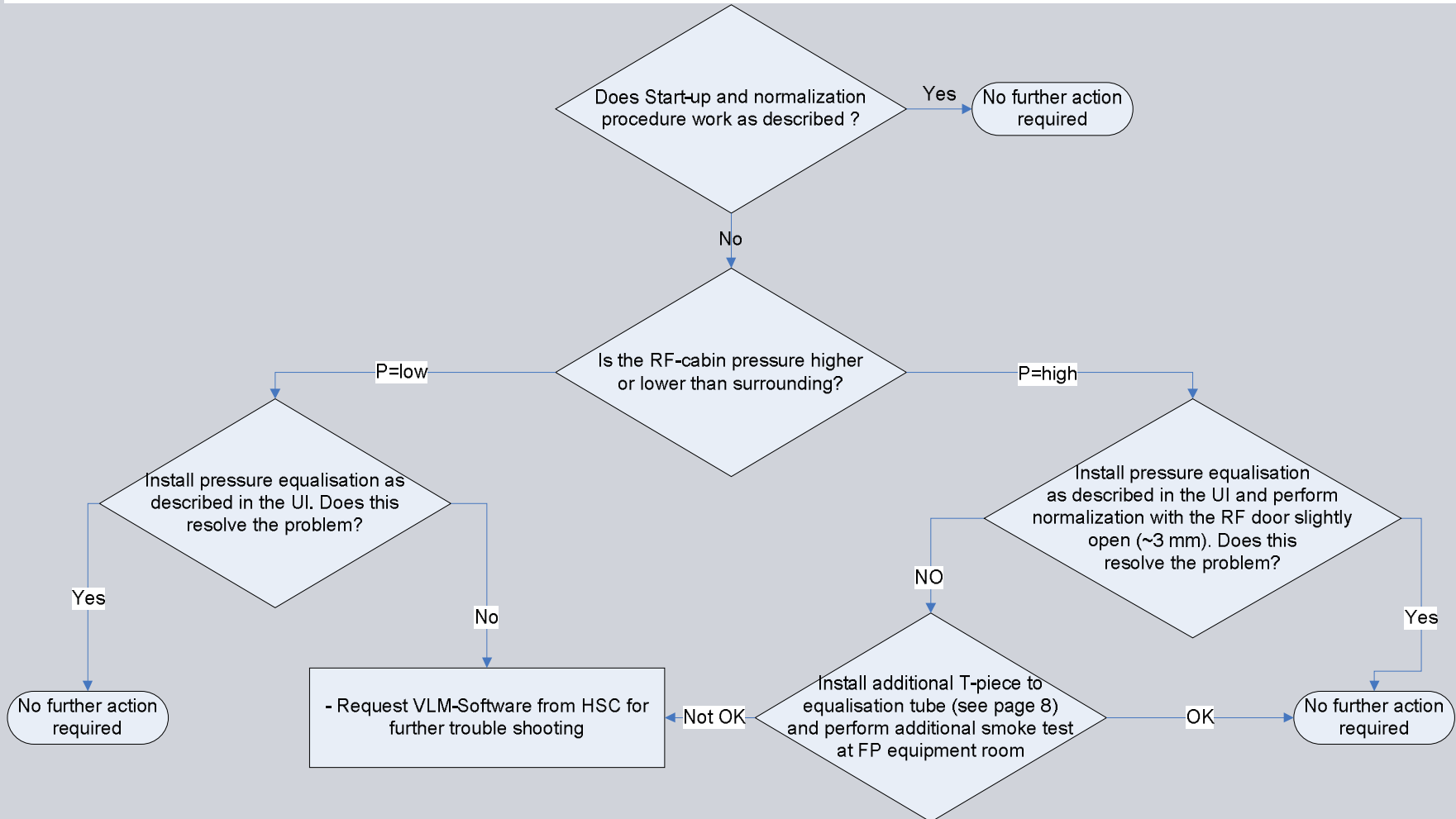
4. Missing sealing in joint elements.
In case of a FAULT condition, check the joint elements (T-pieces) for proper sealing. There must be a white rubber seal as shown on the left in each joint. A missing seal results easily in an air flow difference of 1.4 ltrs./min. (~7%).



5. The black bushing must always be used when a hose is being connected to the inlet or outlet of the VLM. If missing, the flow will increase dramatically for 9 ltrs./min. (~43%). When the grey outlet cover is being removed, the flow increases for 0.4 ltrs./min. (~2%)

For internal use only

How to proceed, when air-pressure related problems occur during VLM start-up



Installation of an additional T-piece in the pressure equalisation hose



Perform a normalisation with the RF cabin door slightly open (~3 mm) first. This results in a normalisation at a mean cabin air pressure between maximum (door closed) and minimum (door open) and gives you a maximum of tolerance from normalised value in both directions.

If this does not resolve the problem, an additional T-piece looped into the pressure equalisation hose may help. Cut the pressure equalisation hose about 1 m behind the VLM exhaust (see left pic.)

Installation of an additional T-piece in the pressure equalisation hose



Prepare a T-piece with appr. 2 m of hose as shown on the left.

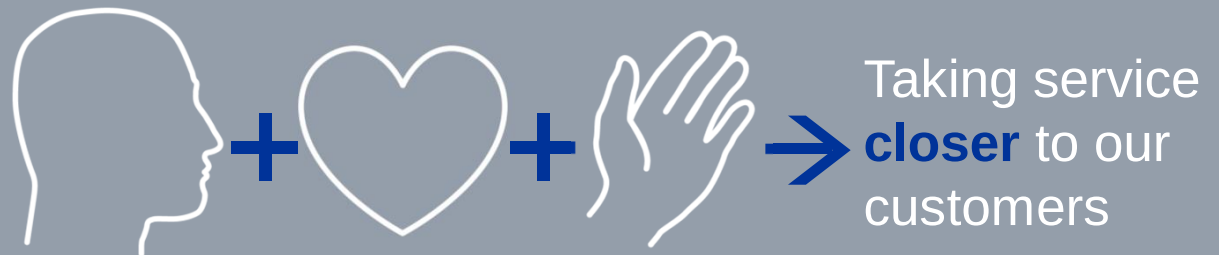
The T-piece can be ordered as a standard spare part with part no.: 10683922 (this T-piece has an additional reduction to 10 mm fitted, which must be removed).

If needed, the black bushing at the VLM-in/outlet can be also ordered as a spare part: 10683929.

Connect the T-piece in line of the previously cut pressure equalisation hose as shown on the left.

> The outlet of the 2 m hose must remain inside the equipment room.





Think. Feel. Do. **Make the difference.**

Contact

Wolfgang Kühnast

H CX CS PS MR

Allee am Röthelheimpark 6
91052 Erlangen

Phone: +49 (9131) 84-7663

Fax: +49 (9131) 84-4721

Mobile: +49 (1522) 2 797793

E-mail: wolfgang.kuehnast@siemens.com